

## Highlights

### **A Reproducible Computational Framework for Fermion Pair Production in Pulsed Electric Fields**

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- Evolves Dirac modes and quantum-kinetic equations independently.
- Benchmarks both solvers against the exact temporal-Sauter spectrum.
- Reports norm, invariant, pulse-tail, and cross-formulation diagnostics.
- Maps absolute finite-window error over momentum and tail extent.
- Includes tested code, reproduction scripts, figures, and a notebook.

# A Reproducible Computational Framework for Fermion Pair Production in Pulsed Electric Fields

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## ARTICLE INFO

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## ABSTRACT

We present PAIRPULSE, a reproducible computational program for momentum-resolved fermion pair production in prescribed, spatially homogeneous electric fields in one spatial dimension. For each canonical momentum, it independently evolves a two-component Dirac mode and the equivalent non-Markovian quantum-kinetic system. The exact asymptotic occupation for a temporal Sauter pulse supplies an external benchmark, while the Dirac norm and quantum-kinetic Bloch-vector invariant provide independent integration diagnostics. The program supports superpositions of temporally displaced Sauter pulses and user-defined pulse objects through a compact field-potential interface. For  $E_0/m^2 = 0.3$  and  $m\tau = 2$ , both solvers reproduce the analytic occupation on 41 momentum points with maximum absolute error below  $8 \times 10^{-13}$ . The reported validation studies comprise 1,962 mode evaluations that separate integration, endpoint and momentum-grid errors and assess parameter and pulse-profile changes. Nine Sauter parameter pairs give maximum absolute occupation error  $4.58 \times 10^{-15}$  at refined settings. Momentum-grid and domain refinements meet a declared  $10^{-4}$  relative density criterion. A momentum-window map shows the absolute finite-window discrepancy across the benchmark grid. A separated two-pulse calculation demonstrates the workflow for a field for which the package supplies no closed-form spectrum. The contribution is a transparent reference implementation with tests and data- and figure-generation scripts; no new pair-production mechanism or analytic result is claimed.

## Program summary

**Program title:** PAIRPULSE

**CPC library link to program files:** Not yet assigned.

**Developer's repository link:** <https://github.com/ozalloum/PairPulse>.

**Code Ocean capsule:** Not applicable.

**Licensing provisions:** MIT License.

**Programming language:** Python 3.10 or later.

**External routines/libraries:** NumPy and SciPy; Matplotlib for plots.

**Nature of problem:** Momentum-resolved fermion pair occupation in a prescribed, spatially homogeneous, time-dependent electric field in 1+1 dimensions.

**Solution method:** Adaptive DOP853 integration of two equivalent but independently implemented state systems: the two-component Dirac equation and the non-Markovian quantum-kinetic equations. The exact temporal-Sauter solution, Dirac norm, quantum-kinetic invariant, and cross-formulation agreement are used for validation.

**Restrictions:** One fermion species in one spatial dimension. Spatial gradients, transverse momentum, collisions, back-reaction, photon dynamics, and self-consistent Maxwell evolution are not included.

**Unusual features:** The exact benchmark and two

state representations are combined with independent conservation checks. Scripts regenerate the manuscript figures, tables, and run metadata.

## 1. Introduction

Nonperturbative pair creation by electric fields is a foundational prediction of quantum electrodynamics. Its early development includes the effective-action treatment of vacuum polarization, Dirac-state transmission in a uniform field, and pair creation in constant or alternating backgrounds [1, 2, 3, 4]. Time-dependent mean-field and kinetic descriptions subsequently clarified how pair occupations evolve, how Pauli blocking enters, and when memory effects matter [5, 6, 7, 8, 9].

For pulsed fields, the final momentum spectrum depends on the full time history of the background. Non-Markovian quantum-kinetic calculations have been compared with scattering and effective-action approaches, and the temporal Sauter field provides an especially useful exact reference case [10, 11, 12, 13, 14, 15]. Short pulses and subcycle structure produce characteristic momentum signatures, while semiclassical turning-point analyses explain interference patterns in time-dependent backgrounds [16, 17, 18, 19, 20, 21]. These studies make clear why a numerical implementation should specify its in/out convention, integration window, and momentum variable rather than report a spectrum without those choices.

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This paper introduces PAIRPULSE, a compact computational-physics program that evolves two independently represented mode systems under the same prescribed field: a two-component Dirac spinor and the non-Markovian quantum-kinetic variables. The exact asymptotic Sauter spectrum provides an external benchmark; conservation diagnostics test each integration internally; and a two-pulse example checks cross-formulation agreement beyond the exactly soluble case. The package includes the source, automated tests, a reproduction notebook, and generated figures. Its scripts generate machine-readable data and run metadata. It is intended as a transparent reference implementation for method checks and modest parameter studies in the spatially homogeneous model, and it does not claim new pair-production physics.

PairPulse's software contribution is the integration of paired Dirac and full auxiliary-variable kinetic evolution, a common in/out convention, an exact Sauter benchmark, conservation diagnostics, and reproducible spectra and density checks in a single computational workflow. This design allows users to check a calculation across formulations and to assess integration, endpoint, and momentum-discretization errors separately.

The remainder of the paper defines the field and mode conventions, describes the software and numerical method, reports validation results, and states the scope and limitations of the program.

## 2. Physical model

We use natural units,  $\hbar = c = 1$ , and temporal gauge,  $A^0 = 0$ . The positive charge magnitude is denoted by  $q$ ; reversing the charge is equivalent to reflecting canonical momentum for the backgrounds considered here. For a spatially uniform vector potential  $A(t)$ , the kinetic momentum of a mode with canonical momentum  $p$  is

$$\pi_p(t) = p - qA(t), \quad E(t) = -\dot{A}(t). \quad (1)$$

The default pulse is

$$\begin{aligned} E(t) &= E_0 \operatorname{sech}^2\left(\frac{t-t_c}{\tau}\right), \\ A(t) &= -E_0\tau \tanh\left(\frac{t-t_c}{\tau}\right). \end{aligned} \quad (2)$$

with  $E_0 \geq 0$ . A pulse train is formed by summing the fields and the corresponding potentials. This preserves Eq. (1) without numerical integration of the electric field.

### 2.1. Dirac-mode evolution

After reduction to a two-component mode, the Hamiltonian can be written

$$H_p(t) = \pi_p(t)\sigma_1 + m\sigma_3, \quad i\partial_t\psi_p(t) = H_p(t)\psi_p(t), \quad (3)$$

where  $m > 0$  and  $\sigma_i$  are Pauli matrices. At the initial and final times, where the pulse tails are small, the instantaneous negative- and positive-energy eigenvectors define the in-vacuum and the out-particle state. The occupation is

$$f_p = \left| u_{p,+}^\dagger(t_f)\psi_p(t_f) \right|^2, \quad (4)$$

with  $\psi_p(t_i) = u_{p,-}(t_i)$ . Exact evolution preserves  $\psi_p^\dagger\psi_p = 1$ .

### 2.2. Quantum-kinetic evolution

The second formulation evolves the distribution  $f_p$  and two auxiliary functions  $u_p, v_p$ . Defining  $\omega_p(t) = \sqrt{m^2 + \pi_p(t)^2}$  and  $W_p(t) = qE(t)m/\omega_p(t)^2$ , the equations are

$$\begin{aligned} \dot{f}_p &= \frac{1}{2}W_p u_p, \\ \dot{u}_p &= W_p(1 - 2f_p) - 2\omega_p v_p, \\ \dot{v}_p &= 2\omega_p u_p, \end{aligned} \quad (5)$$

with  $(f_p, u_p, v_p) = (0, 0, 0)$  at the finite initial time  $t_i$ . This represents the instantaneous vacuum used by the Dirac solver, and approaches the asymptotic in-vacuum as the pulse tails vanish. The local auxiliary-variable system represents the non-Markovian kinetic evolution [8, 12]. These equations preserve

$$(1 - 2f_p)^2 + u_p^2 + v_p^2 = 1, \quad (6)$$

which is used as an independent integration diagnostic. The asymptotic value of  $f_p$  is compared directly with Eq. (4).

### 2.3. Exact Sauter benchmark

For Eq. (2), let  $\lambda = qE_0\tau^2$  and

$$\omega_\pm = \sqrt{m^2 + (p \pm qE_0\tau)^2}. \quad (7)$$

The asymptotic occupation per fermionic mode, with no additional spin-degeneracy factor, is [13]

$$f_p^{\text{exact}} = \frac{\cosh(2\pi\lambda) - \cosh[\pi\tau(\omega_+ - \omega_-)]}{2\sinh(\pi\tau\omega_+)\sinh(\pi\tau\omega_-)}. \quad (8)$$

The spin-summed distribution in Ref. [13], Eq. (99), is twice the per-mode occupation used here, with transverse momentum set to zero. This normalization is consistent with the single-species density below. The implementation evaluates an algebraically equivalent logarithmic expression to reduce overflow in the hyperbolic functions. This algebraic reformulation does not guarantee relative accuracy for arbitrarily small occupations. Equation (8) is used only for a single Sauter pulse; pulse-train spectra are computed by time evolution.

### 3. Software and numerical method

PAIRPULSE is a Python package built around a small pulse interface exposing  $E(t)$ ,  $A(t)$ , and a finite integration window. It provides an analytic Sauter pulse and finite pulse trains, but the solvers accept user-defined pulse objects implementing the same interface. This separates the field model from the mode evolution and lets users replace the pulse without changing either solver.

Both initial-value problems use SciPy's DOP853 solver through `solve_ivp`. The default relative and absolute tolerances are  $2 \times 10^{-10}$  and  $2 \times 10^{-12}$ . The integration window extends 12 duration parameters beyond the outermost pulse centers by default. For each mode, 257 equally spaced times in  $[t_i, t_f]$  are used to estimate  $\omega_{\max} = \max \sqrt{m^2 + [p - qA(t)]^2}$ , and the maximum internal step is  $c_h/\omega_{\max}$ , where the public parameter `max_step_coefficient` defaults to  $c_h = 0.18$ . This sampled cap is empirical; it is not a rigorous frequency bound for an arbitrary user-defined pulse. The window length is user-adjustable; the manuscript's tail study makes its effect explicit. The same time interval, DOP853 implementation, tolerances, and step-cap rule are used for both formulations. Thus the state equations are implemented independently, but their time integrator is shared. Agreement between them does not by itself exclude common integration or window errors. Custom pulse objects must supply mutually consistent  $E(t)$  and  $A(t)$  with sufficiently decayed endpoint fields; the interface does not check these conditions automatically.

For a single Sauter pulse, PAIRPULSE reports the exact occupation, direct Dirac occupation, quantum-kinetic occupation, final-time norm/invariant errors, and ODE function-evaluation counts. These diagnostics are not maxima over the intermediate time history. The two reproduction scripts regenerate the six manuscript figures, comma-separated tables, and environment metadata. The additional `redraw_figures.py` script redraws the figures from CSVs generated by the reproduction scripts without repeating mode integrations. The accompanying notebook shows the central benchmark and pulse-train example. The eight tests cover the exact benchmark, the quantum-kinetic invariant and occupation bounds, the zero-field limit, pulse superposition, time translation, step-cap refinement, and rejection of invalid controls or nonfinite pulse parameters.

Figures 4 and 5 summarize the software organization, generated outputs, and the sequence used to regenerate datasets and publication figures. Both diagrams are generated by the included Python plotting script. The separate finite-window scan writes its underlying momentum-tail-extent values to a CSV file, in addition to the spectra, convergence data, and run metadata produced by the main reproduction script.

The one-dimensional momentum-integrated density is defined as

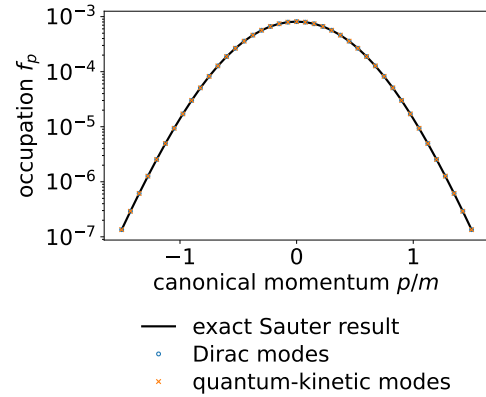
$$n_{1D} = \int \frac{dp}{2\pi} f_p, \quad (9)$$

and is evaluated with the trapezoidal rule on the supplied finite momentum grid. It is a one-dimensional density for one mode species, not a three-dimensional number density. Users must check momentum-window and grid convergence for applications beyond the examples below.

## 4. Results

### 4.1. Single Sauter pulse

The reference case uses  $m = q = 1$ ,  $E_0 = 0.3$ ,  $\tau = 2$ , and 41 equally spaced canonical momenta on  $[-1.5, 1.5]$ . Figure 1 compares both numerical formulations with Eq. (8). Across this grid, the largest absolute errors are  $7.48 \times 10^{-13}$  for the Dirac modes and  $7.54 \times 10^{-13}$  for the quantum-kinetic calculation. The maximum relative errors over modes with  $f_p^{\text{exact}} > 10^{-12}$  are  $1.68 \times 10^{-8}$  and  $1.68 \times 10^{-8}$ , respectively. The maximum final-time Dirac norm error across the momentum grid is  $6.24 \times 10^{-13}$ ; the maximum final-time deviation of the quantum-kinetic invariant from unity is  $9.13 \times 10^{-13}$ .



**Figure 1:** Single-pulse occupation spectrum for  $E_0 = 0.3$ ,  $\tau = 2$ ,  $t_c = 0$ , and  $m = q = 1$ , evaluated at 41 equally spaced canonical momenta in  $[-1.5, 1.5]$ . The black line is the exact asymptotic Sauter result; circles and crosses denote Dirac and quantum-kinetic modes integrated over  $[-12\tau, 12\tau]$ . The numerical results lie on the exact curve at this resolution. The vertical axis is logarithmic; markers represent deterministic calculations, with no statistical error bars.

At the fixed solver settings used here, short integration windows give larger discrepancies from the asymptotic benchmark. At  $p = 0$  the exact value is  $8.11991615993 \times 10^{-4}$ . Table 1 shows that extending the window from six to twelve pulse durations reduces the absolute error by more than five orders of magnitude. At  $14\tau$ , discrepancies are of order  $10^{-14}$ . This scan

**Table 1**

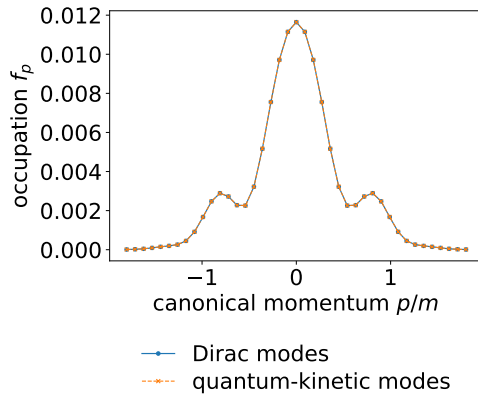
Finite-window convergence at  $p = 0$  for  $E_0 = 0.3$ ,  $\tau = 2$ ,  $t_c = 0$ , and  $m = q = 1$ . Endpoints are  $t_i = -L$ ,  $t_f = L$ . Both solvers use relative tolerance  $2 \times 10^{-10}$ , absolute tolerance  $2 \times 10^{-12}$ , and the step cap in Section 3. Errors are absolute differences from Eq. (8); numerical tolerances are held fixed as  $L$  varies.

| Tail half-width $L/\tau$ | Exact                          | Dirac absolute error     | QKE absolute error       |
|--------------------------|--------------------------------|--------------------------|--------------------------|
| 6                        | $8.11991615993 \times 10^{-4}$ | $1.1762 \times 10^{-7}$  | $1.1762 \times 10^{-7}$  |
| 8                        | $8.11991615993 \times 10^{-4}$ | $2.1989 \times 10^{-9}$  | $2.1989 \times 10^{-9}$  |
| 10                       | $8.11991615993 \times 10^{-4}$ | $4.0728 \times 10^{-11}$ | $4.0721 \times 10^{-11}$ |
| 12                       | $8.11991615993 \times 10^{-4}$ | $7.4813 \times 10^{-13}$ | $7.5435 \times 10^{-13}$ |
| 14                       | $8.11991615993 \times 10^{-4}$ | $1.2825 \times 10^{-14}$ | $6.7149 \times 10^{-15}$ |

alone cannot separate residual tail error from integration error and roundoff; Section 4.4 reports separate refinements.

## 4.2. Separated pulses

As a case for which the package supplies no closed-form benchmark, we superpose two same-polarity Sauter pulses, each with  $E_0 = 0.3$  and  $\tau = 1.1$ , centered at  $t_c = \pm 3.5$ . Figure 2 shows the computed momentum spectrum on  $[-1.8, 1.8]$ . The two independent state representations agree over the full grid; both resolve shoulders and modulations away from the central maximum. The spectrum is included as a cross-formulation code demonstration. Its detailed features are not assigned to a specific interference mechanism on the basis of this calculation alone.



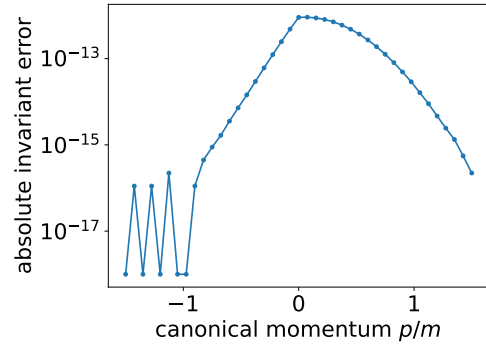
**Figure 2:** Occupation spectrum for two separated, same-polarity Sauter pulses, each with  $E_0 = 0.3$  and  $\tau = 1.1$ , centered at  $t_c = \pm 3.5$ , with  $m = q = 1$ . Circles with a solid line and crosses with a dashed line denote Dirac and quantum-kinetic calculations on 41 equally spaced canonical momenta in  $[-1.8, 1.8]$ . Endpoints are  $t_i = -16.7$ ,  $t_f = 16.7$ . Lines join computed modes; no analytic curve is supplied for this case.

The largest absolute difference between the Dirac and quantum-kinetic occupations on this grid is  $1.18 \times 10^{-13}$ . For orientation, trapezoidal integration over the stated finite windows gives  $n_{1D} = 1.13951 \times 10^{-4}$  for the single-pulse exact spectrum and  $1.69199 \times 10^{-3}$  for the double-pulse spectrum. These original values use the

finite grids shown in the figures. Section 4.6 separately refines the momentum spacing and domain.

## 4.3. Evolution diagnostics

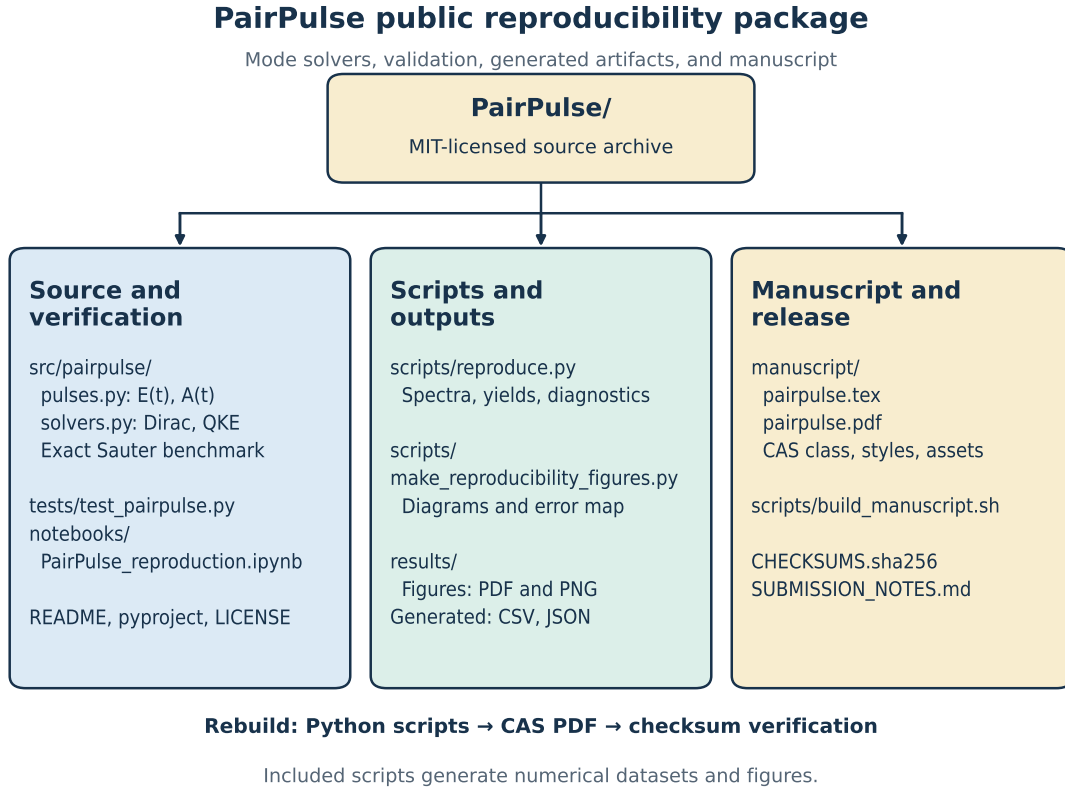
Figure 3 displays the absolute quantum-kinetic invariant error for the single-pulse momentum grid. The maximum error is below  $10^{-12}$  in this run. The invariant is a useful internal diagnostic, but it does not replace comparison with the exact occupation or a convergence study in momentum and time.



**Figure 3:** Final-time quantum-kinetic invariant error  $|(1 - 2f_p)^2 + v_p^2 + v_p^2 - 1|$  for the single-pulse setup of Figure 1. Each marker corresponds to one of 41 momentum modes, evaluated at  $t_f = 12\tau$ ; the connecting line guides the eye. Values are plotted on a logarithmic scale with a display floor of  $10^{-18}$ . The diagnostic measures conservation of the invariant, not occupation error or a maximum over time.

Short integration windows can shift the computed spectrum away from its asymptotic value. Figure 6 shows the absolute error for both formulations. The scan uses 41 momenta and six tail half-widths, with  $L/\tau$  from 4 to 14 in steps of 2. Parameters are  $E_0/m^2 = 0.3$  and  $m\tau = 2$ .

Here  $L$  is the half-width from the pulse center to either integration endpoint. The plotted discrepancy includes finite-window truncation and fixed-tolerance integration error; small discrepancies are not necessarily pure tail error. The CSV records both occupations, the exact result, errors, and solver diagnostics. Because the field is spatially homogeneous, the map axes are momentum and window extent.



**Figure 4:** Organization of PAIRPULSE and its generated outputs. Source modules, tests, notebook, and run scripts are shown alongside the generated datasets, figures, manuscript, and release checks. The diagram is produced in Python from the package's declared file structure.

#### 4.4. Validation protocol and integration refinement

The validation studies reported here comprise 1,962 individual mode evaluations, all completed without integration failures. The protocol was recorded before execution. Its acceptance criteria are occupation differences below  $10^{-10}$ , relative errors below  $10^{-5}$  when the exact occupation is at least  $10^{-10}$ , successive relative changes below  $10^{-4}$  for the integrated density, and no occupation outside  $[0, 1]$  by more than  $10^{-10}$ . These are empirical acceptance criteria, not certified error bounds. The scripts retain unsuccessful integrations rather than silently discarding them.

At fixed tail extent 20, each of the single- and double-pulse cases is tested at  $p = 0$ ,  $p = \pm 0.75$  and  $p = \pm 1.5$ . The relative tolerance takes values  $2 \times 10^{-8}$ ,  $2 \times 10^{-10}$  and  $2 \times 10^{-12}$ , with absolute tolerance one hundredth as large; independently,  $c_h$  takes values 0.18, 0.09 and 0.045. Both formulations are run at all combinations (180 evaluations). The largest occupation difference from the same formulation at the tightest tolerance and cap is  $8.46 \times 10^{-14}$ . Refinement differences need not decrease monotonically once errors approach floating-point precision. The refined double-pulse solution is a numerical reference, not an exact solution.

Separately, 60 evaluations vary the tail extent through 12, 16 and 20 at relative tolerance  $2 \times 10^{-12}$  and  $c_h = 0.045$ . Relative to extent 20, the largest occupation differences at extent 12 are  $7.48 \times 10^{-13}$  for the single pulse and  $2.59 \times 10^{-13}$  for the double pulse; at extent 16 they are below  $2.51 \times 10^{-16}$  and  $9.37 \times 10^{-16}$ , respectively. This distinguishes endpoint sensitivity from the fixed-window integration study.

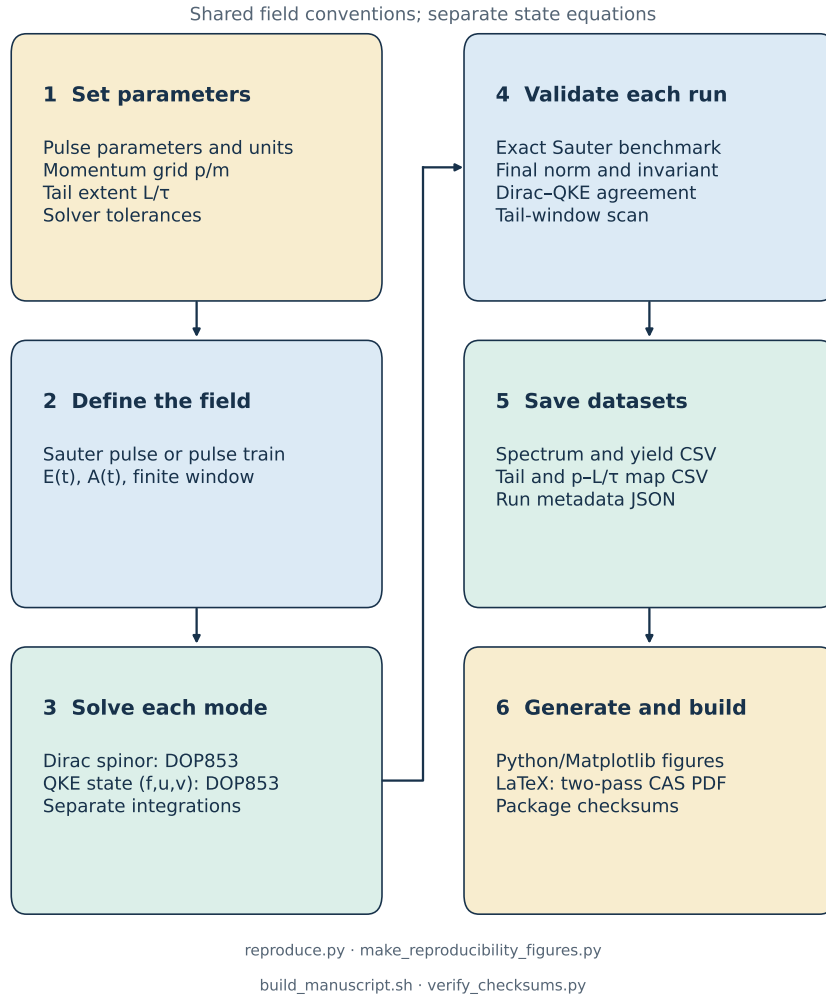
#### 4.5. Parameter coverage and a custom pulse

The Sauter benchmark is evaluated at all nine combinations of  $E_0 = 0.1, 0.3, 0.8$  and  $\tau = 0.5, 2, 5$ , with  $m = q = 1$ . Each uses 13 equally spaced momenta on  $[-p_{\max}, p_{\max}]$ , where  $p_{\max} = \max(1.5, E_0\tau + 2/\tau)$ , tail extent 20, relative tolerance  $2 \times 10^{-12}$ , absolute tolerance  $2 \times 10^{-14}$  and  $c_h = 0.09$ . Table 2 reports all 234 evaluations. The largest absolute error is  $4.58 \times 10^{-15}$ , and the largest relative error over resolved occupations is  $1.83 \times 10^{-9}$ . No negative occupations or upper-bound violations occur in this sweep. Exact occupations reach  $7.81 \times 10^{-19}$ ; relative accuracy is not asserted for these smallest values.

A user-defined Gaussian profile provides an additional interface check:

$$E(t) = 0.3 \exp[-(t/2)^2],$$

## Computational workflow



**Figure 5:** Computational workflow used to generate the reproducibility datasets and publication figures. Each momentum mode is evolved independently with the Dirac and quantum-kinetic systems; the exact Sauter result and conservation diagnostics are then used where applicable before data export, plotting, and manuscript compilation.

**Table 2**

Maximum absolute Sauter occupation error over 13 momenta per parameter pair, for each formulation. Settings are specified in the text.

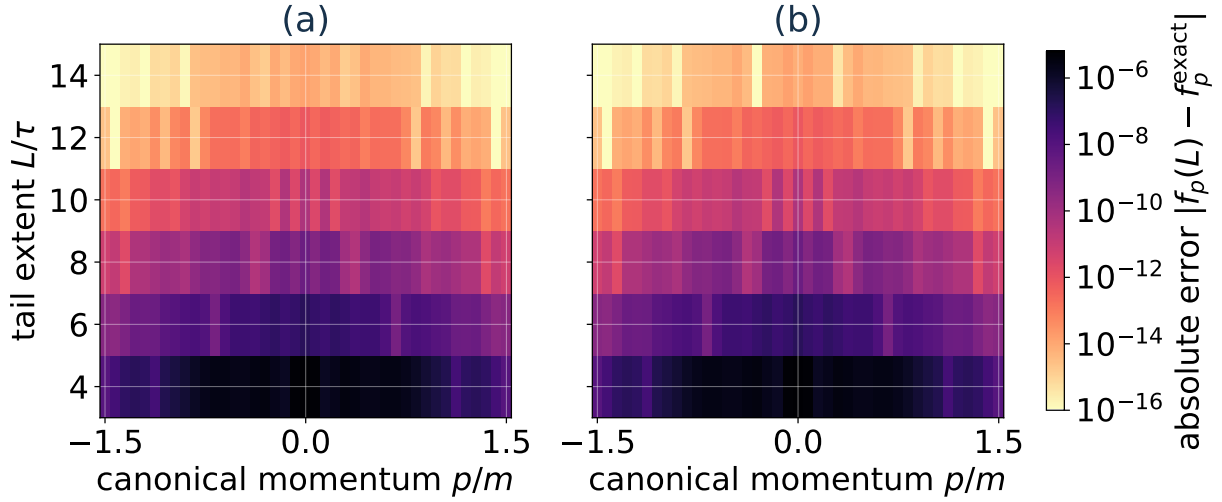
| $E_0$ | $\tau$ | Dirac                  | QKE                    |
|-------|--------|------------------------|------------------------|
| 0.1   | 0.5    | $4.84 \times 10^{-17}$ | $8.19 \times 10^{-16}$ |
| 0.1   | 2      | $4.02 \times 10^{-18}$ | $1.71 \times 10^{-18}$ |
| 0.1   | 5      | $8.39 \times 10^{-20}$ | $4.12 \times 10^{-19}$ |
| 0.3   | 0.5    | $9.71 \times 10^{-17}$ | $2.71 \times 10^{-15}$ |
| 0.3   | 2      | $5.43 \times 10^{-17}$ | $2.76 \times 10^{-17}$ |
| 0.3   | 5      | $5.90 \times 10^{-17}$ | $9.67 \times 10^{-18}$ |
| 0.8   | 0.5    | $4.27 \times 10^{-15}$ | $4.58 \times 10^{-15}$ |
| 0.8   | 2      | $6.45 \times 10^{-16}$ | $1.77 \times 10^{-16}$ |
| 0.8   | 5      | $1.54 \times 10^{-15}$ | $1.42 \times 10^{-16}$ |

$$A(t) = -0.3\sqrt{\pi} \operatorname{erf}(t/2). \quad (10)$$

These expressions satisfy  $E = -\dot{A}$ . Five momenta ( $p = 0, \pm 0.75, \pm 1.5$ ), two endpoint extents ( $|t| = 12, 16$ ), two cap coefficients (0.09, 0.045) and the two PairPulse solvers give 40 evaluations at relative tolerance  $2 \times 10^{-12}$ . The largest absolute difference between Dirac and quantum-kinetic occupations at matching settings is  $1.31 \times 10^{-17}$ . This checks cross-formulation agreement for one smooth custom profile; it does not establish absolute accuracy or validate arbitrary user-supplied fields.

### 4.6. Momentum-grid and domain refinement

The QKE density is first evaluated at 41, 81 and 161 nodes on the original fixed momentum interval. The spacing of the 161-node grid is then held constant as the interval is doubled and tripled, using 321 and 481 nodes. All modes are evaluated directly, without



**Figure 6:** Absolute finite-window discrepancy  $|f_p(L) - f_p^{\text{exact}}|$  from the asymptotic Sauter benchmark: (a) Dirac modes and (b) quantum-kinetic modes, for  $E_0 = 0.3$ ,  $\tau = 2$ ,  $t_c = 0$ , and  $m = q = 1$ . Each panel contains 41 momenta in  $[-1.5, 1.5]$  and six half-widths  $L/\tau = 4, 6, 8, 10, 12, 14$ ; integration endpoints are  $t_c \pm L$ . The common logarithmic color scale uses a display floor of  $10^{-16}$  without changing the computed errors. Discrepancies include window, integration, and roundoff effects at fixed tolerances. The horizontal coordinate is canonical momentum, not position.

imposing spectrum symmetry. A nested 481-node grid per case supplies the 962 mode evaluations in this study. Tail extent 20, relative tolerance  $2 \times 10^{-12}$ , absolute tolerance  $2 \times 10^{-14}$  and  $c_h = 0.09$  are used throughout.

**Table 3**

Trapezoidal density refinement.  $P$  is the half-width of the momentum interval and  $N_p$  is its number of nodes. All values use QKE.

| Case   | $P$ | $N_p$ | $n_{1D}$                      |
|--------|-----|-------|-------------------------------|
| Single | 1.5 | 41    | $1.1395114844 \times 10^{-4}$ |
|        | 1.5 | 81    | $1.1395130337 \times 10^{-4}$ |
|        | 1.5 | 161   | $1.1395134244 \times 10^{-4}$ |
|        | 3.0 | 321   | $1.1395539392 \times 10^{-4}$ |
|        | 4.5 | 481   | $1.1395539392 \times 10^{-4}$ |
| Double | 1.8 | 41    | $1.6919881473 \times 10^{-3}$ |
|        | 1.8 | 81    | $1.6919976391 \times 10^{-3}$ |
|        | 1.8 | 161   | $1.6920000711 \times 10^{-3}$ |
|        | 3.6 | 321   | $1.6926588379 \times 10^{-3}$ |
|        | 5.4 | 481   | $1.6926588400 \times 10^{-3}$ |

The last fixed-domain grid refinement changes the single- and double-pulse densities by  $3.43 \times 10^{-7}$  and  $1.44 \times 10^{-6}$  relatively. The final domain extension changes them by  $7.20 \times 10^{-13}$  and  $1.21 \times 10^{-9}$ , respectively. The largest difference between the QKE and analytic single-pulse densities on identical grids is below  $1.71 \times 10^{-18}$ . A further 486 mode evaluations on 81-node grids vary tail extents (12, 16, 20) and halve the cap coefficient at extent 20; the largest relative density change is  $4.98 \times 10^{-11}$ . The density checks therefore meet the declared  $10^{-4}$  criterion, while remaining finite

refinement evidence rather than a rigorous continuum error estimate.

#### 4.7. Installation and notebook execution

The reported numerical studies were executed on 8 October 2026 with Python 3.12.14, NumPy 2.3.5 and SciPy 1.17.0. A built wheel was installed into two isolated Windows environments: Python 3.12.14 with the reproduction libraries, and Python 3.10.20 with NumPy 1.24.0, SciPy 1.10.0 and Matplotlib 3.7.0. All eight tests passed in both environments. Separate hosted checks successfully built and installed the package from source on Linux and Windows. On both systems, the notebook executed all four code cells in a real Jupyter kernel, imported the installed package, and produced no error outputs.

#### 5. Limitations and intended use

The implementation assumes a prescribed, spatially homogeneous background and one spatial dimension. It does not include back-reaction, collisions, spatial gradients, transverse momentum, scalar particles, or a self-consistent electromagnetic field. PAIRPULSE is intended for calculations within these stated physical assumptions.

The exact Sauter formula validates asymptotic occupation for a single pulse; it does not validate every user-supplied field. For new profiles, users should vary tolerances, pulse-tail windows, time-step limits, and momentum resolution, and compare physical observables with an independent method where possible. The reported occupations are evaluated at the specified finite integration endpoints.

The studies in Section 4 separate integration controls, endpoint extent, momentum spacing and momentum range, and cover nine single-pulse parameter pairs plus a Gaussian profile. This finite set does not establish a parameter-independent accuracy bound. Very weak occupations, stronger or longer pulses, sharply structured custom fields and larger pulse trains require further checks.

The code emphasizes clarity and cross-validation over throughput. Each momentum mode is integrated independently in Python, so the present package is most appropriate for reference calculations, method checks, and modest parameter sweeps. The validation script can distribute independent modes among subprocess workers. Compiled kernels and demonstrated scaling for large parameter scans remain outside the current assessment.

## 6. Conclusion

PAIRPULSE provides an inspectable implementation of two established formulations for fermion pair creation in homogeneous pulsed electric fields. The exact Sauter solution, norm and invariant diagnostics, pulse-tail data, separated-pulse example, and executable notebook make the numerical conventions reproducible. The two numerical formulations reproduce the exact benchmark to below  $8 \times 10^{-13}$  absolute error for the stated parameter set and agree to about  $1.2 \times 10^{-13}$  for the two-pulse example. The additional 1,962 evaluations reported here separate integration and domain errors, cover nine Sauter parameter pairs, and test cross-formulation agreement for a custom Gaussian pulse. All declared numerical criteria pass. These results validate the implementation for the tested cases; they do not establish a general accuracy guarantee or claim new pair-production physics.

The deliberately limited model makes the package useful as a reference for homogeneous-field calculations and as a base for extensions. Applications involving spatial dependence, back-reaction, or additional particle degrees of freedom require further derivations and validation against appropriate independent methods.

## Code and data availability

The PAIRPULSE source code, reproduction scripts, notebook, manuscript, and generated figures are available in the versioned release v0.2.1 at <https://github.com/ozalloum/PairPulse/releases/tag/v0.2.1>. This release and the matching submission package do not include numerical validation records, CSV datasets, executed notebooks, or execution logs. The included scripts generate numerical datasets and diagnostics from fresh calculations; installation and reproduction instructions are provided in the repository README. The PairPulse code is distributed under the MIT License; the bundled Elsevier CAS files retain

their upstream license notices. The release tag identifies the corresponding source commit, and SHA-256 manifests identify the distributed files. No archival DOI is claimed.

## CRediT authorship contribution statement

Othman H. Y. Zalloum: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Visualization, Project administration, Writing—original draft, Writing—review and editing.

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## Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

ChatGPT (OpenAI) was used to assist with manuscript editing, literature and reference checks, source-code inspection, plotting-script revisions, reproducibility checks, and LaTeX formatting. The scientific plots were generated with Python from numerical calculations or archived data. The corresponding author is responsible for reviewing and verifying the scientific content, computations, references, and final text before submission.

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